MIL-L-45946A
17 August 1979
SUPERSEDING
MIL-L-45946
31 July 1972

MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC BELT, 20MM: M22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of metallic belt link intended to hold 20 millimeter (mm) ammunition in the form of flexible belts for use in unloading linkless feed systems.

2. APPLICABLE DOCUMENTS

2.1 The following, of the issue in effect on date of invitation for bids or request for proposals, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-13855	-	Weapons, Small Arms, General Specification for
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-16232	-	Phosphate, Coating, Heavy, Manganese or Zinc Base

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS (See 6.3)

U.S.ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

8448023 - Link, Cartridge, Metallic Belt, 20MM: M22
8448023 - Packaging Data Sheet for M22 Link, Cartridge, Metallic Belt, 20MM: M22
11691592 - Requirement, Test Diagram for Link, Cartridge, Metallic Belt, 20MM: M22

INSPECTION EQUIPMENT DRAWINGS

7190394 - Progressive Snap Gage
8409005 - Loading Fixture
8440919 - Gage, Dial Indicating
8441498 - Gage, Flush Pin
11838145 - Fixture, Test, Pull
11838147 - Gage, Plug, Flat
11838231 - Fixture, Test, Extraction
11838232 - Pump Assy, Hydraulic, Electric Powered
11838277 - Set-Up and Operating Instructions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E18 - Test for Rockwell Hardness and Rockwell Superfical Hardness of Metallic Materials
ASTM E112 - Methods for Estimating Average Grain Size of Metals
ASTM B117 - Method of Test for Salt Spray (Fog) Testing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. - The links shall comply with all requirments specified on drawings 8448023 and 11691592 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Delinking. - A linked M51 series 20MM dummy cartridge shall be capable of being disconnected and reconnected by hand from a belt of cartridges without evidence of distortion. The test shall be performed as specified in 4.5.3.

3.4 Functioning. - The links shall be capable of being assembled into belts with M51 series 20MM dummy cartridges and being subjected to the loading cycle of a linkless feed system without separating or causing a malfunction. The test shall be performed as specified in 4.5.8.

3.5 Marking. - The links shall be marked in accordance with MIL-STD-130 and Drawing (Dwg.) 8448023. In addition, the links shall be identified with the inspection lot number as specified in MIL-STD-1168.

3.6 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. - Workmanship shall be in accordance with MIL-W-13855. Cracks and tears in non-principal areas shall not exceed the limits specified in 6.4.2.2 and 6.4.3.2. Deformations shall not exceed the limits specified in 6.4.1. Burrs in non-principal areas shall not exceed the limits specified in 6.4.3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspections shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantitites as indicated:

Part Description	Drawings	Quantity
Link, Cartridge, Metallic Belt, 20MM: M22	8448023	2400

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - Rejection of the first article sample shall be in accordance with Table I and MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45946A

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M22		1 OF 2	See Below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Link, Cartridge, Metallic Belt, 20MM: M22 (Drawing 8448023)				
	Examination for defects	200	ACC-0 REJ-1	3.2, 3.5, 3.7	4.4.2.1 (a)
	Elongation	200	ACC-1 REJ-2	3.2	4.5.2
	Delinking	32	ACC-0 REJ-1	3.3	4.5.3
	Grain Size	5	ACC-0 REJ-1	3.2	4.5.4
	Decarburization	5	ACC-0 REJ-1	3.2	4.5.5
	Salt Spray	8	ACC-0 REJ-1	3.2	4.5.6
	Hardness	50	ACC-0 REJ-1	3.2	4.5.7
	Requirements, Test Diagram for Link, Cartridge, Metallic Belt, 20MM: M22 (Drawing 11691592)				
	Examination for defects	800	ACC-0 REJ-1	3.2	4.4.2.2 (b)

NOTES: (a) Inspection shall be performed for Majors 101 thru 113; and Minors 201 thru 204 of 4.4.2.1.

(b) Inspection shall be performed for Majors 101 and 102 of 4.4.2.2.

Source: https://assist.dla.mil -- Downloaded: 2020-10-22T04:25Z
Check for updates or verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45946A

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20MM: M22	2	2	See Below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge extraction	500	ACC-4 REJ-5	3.2	8409005, 11838231, 11829232, 11838277
	Functioning	600	ACC-0 REJ-1	3.4	4.5.8

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisons of MIL-A-48078 and MIL-W-13855. In addition, inspection lots shall not exceed 500,000 links.

4.4.1.1 Heat treat lot. - The heat treat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one change or in a continuous type of furnace during an uninterrupted period extending over not more than one eight hour shift.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45946A

PARAGRAPH	TITLE		SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Link, Cartridge, Metallic Belt, 20MM:: M22			8448023
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Cracks or tears in principal areas as defined in 6.4.2.1		0.25%	3.2	4.5.1
102.	Burrs in principal areas as defined in 6.4.3.1		0.25%	3.2	Visual
103.	Width of connecting arm		0.25%	3.2	Gage
104.	Width of connecting arm eye		0.25%	3.2	Gage
105.	Width across connecting arm tabs		0.25%	3.2	Gage
106.	Width across connecting arm tabs' radii		0.25%	3.2	Gage
107.	Front guide to rear locating tab		0.65%	3.2	8440919
108.	Overall length of body		0.65%	3.2	Gage
109.	Width of stripper bridge, min.		1.00%	3.2	11838147
110.	Detent to rear locating tab (2)		1.00%	3.2	Gage
111.	Protrusion of end detent		1.50%	3.2	8441498
112.	Height of stripper bridge		1.50%	3.2	7190394
113.	Pitch dimension		1.50%	3.2	11838145
114.	Elongation	200	ACC-2 REJ-3	3.2	4.5.2 / 11838145
115.	Delinking	20	ACC-0 REJ-1	3.3	4.5.3
116.	Grain Size	3(a)	ACC-0 REJ-1	3.2	4.5.4
117.	Decarburization	3(a)	ACC-0 REJ-1	3.2	4.5.5
118.	Salt Spray	(b)	(b)	3.2	4.5.6
119.	Hardness		0.40%(c)	3.2	4.5.7

NOTES: (a) Selected from each heat treat lot (See 4.4.1.1)
(b) Sampling and rejection shall be in accordance with MIL-P-16232.
(c) Inspection Level S-4 of MIL-STD-105. Samples selected from each heat treat lot (See 4.4.1.1)

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45946A

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Link, Cartridge, Metallic Belt, 20MM: M22				8448023
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Incorrect, missing or illegible marking		2.50%	3.5	Visual
202.	Incorrect or missing protective finish		2.50%	3.2	Visual
203.	Evidence of poor workmanship, or cracks, tears or burrs in nonprincipal areas as defined in 6.4.2.2 and 6.4.3.2 or deformations as defined in 6.4.1		4.00%	3.7	Visual/4.5.1
204.	Depth of side detent (2)		4.00%(d)	3.2	Gage

NOTES:
(d) Inspection Level S-3 of MIL-STD-105.

Source: http://assist.dla.mil - Downloaded: 2026-Feb-27 10:04 EST
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11691592
4.4.2.2	Requirements, Test Diagram for Link, Cartridge, Metallic Belt, 20MM: M22				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Height from bottom of dummy cartridge to top of link, max.	800	ACC-0 REJ-1	3.2	Gage
102.	Distance from end of dummy cartridge to front guide of link, max.	800	ACC-0 REJ-1	3.2	Gage
103.	Cartridge extraction	500	ACC-5 REJ-6	3.2	8409005, 11838231, 11838232, 11838277
Minor	None defined				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45946A

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Link, Cartridge, Metallic Belt, 20MM: M22 - Packaging Data Sheet	1 OF 1	8448023
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Improper level of packaging and packing		1.50%	3.2	Visual
202.	Incorrect, missing or illegible markings on containers		1.50%	3.2	Visual
203.	Inadequate cleaning or drying of links		1.50%	3.2	Visual
204.	Unit container interior contaminated with dirt, lint, or other foreign matter		1.50%	3.2	Visual
205.	Improper orientation of links in unit container		1.50%	3.2	Visual
206.	Layers of links not separated properly		1.50%	3.2	Visual
207.	Improper closure of unit container		1.50%	3.2	Visual
208.	Improper packing of unit containers		1.50%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22 04:12:42
Check the source to verify that this is the current version before use.

4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and test prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection

4.5.1 Inspection for cracks or tears. - This inspection shall be performed using 5x magnification. Observation shall be made for compliance with Section 6 herein.

4.5.2 Elongation. - The link shall be inserted into the test fixture and the pitch dimension shall be measured and recorded. The applicable load shall be gradually applied at a uniform rate on the center line of the link. The full load shall then be removed and the pitch dimension shall be remeasured and recorded. Observation shall be made for compliance with the applicable drawing requirement. Links so tested shall be scrapped.

4.5.3 Delinking. - The sample links shall be belted with M51 series 20MM dummy cartridges. The links shall be disconnected and then reconnected by hand without the use of any special equipment or tools. Observation shall be made for compliance with the applicable specification requirement.

4.5.4 Grain size. - Grain size shall be determined by ASTM Method E112. Observation shall be made for compliance with the applicable drawing requirement.

4.5.5 Decarburization. - The cross section of the sample link shall be polished, etched with 3 to 5% Nital solution and examined under a microscope of not less than 500 power. Observation shall be made for compliance with the applicable drawing requirement.

4.5.6 Salt spray. - Salt spray testing shall be conducted in accordance with ASTM Method B117. Observation shall be made for compliance with the applicable requirement in MIL-P-16232.

4.5.7 Hardness. - Hardness testing shall be conducted in accordance with ASTM Method E18. Observation shall be made for compliance with the applicable drawing requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

4.5.8 Functioning. - The sample links shall be tested by the Government by assembling them into belts using M51 series 20MM dummy cartridges and subjecting the belts to a loading cycle of any linkless feed system specified by the testing agency.

5. PACKAGING

5.1 Preservation and packaging. - The links shall be preserved and packaged for shipment in accordance with Packaging Data Sheet P8448023 with levels as specified in the contract (See 6.1).

5.2 Packing. - The links shall be packed in accordance with Packaging Data Sheet P8448023 with levels as specified in the contract (See 6.1).

5.3 Marking. - Marking shall be in accordance with Packaging Data Sheet P8448023.

5.4 Shipping. - When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. - See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Dover, N.J. 07801, ATTN: DRDAR-QAA-I

6.3 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armamemnt Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 Definition of terms

6.4.1 Deformed link. - A link whose shape or dimensions or both have been mutilated or altered by processing to the extent that it is unsuitable for its intended use.

6.4.2 Cracks - tears. - See Figures 1, 2 and 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

6.4.2.1 Cracks - tears in principal areas. - Any visible fracture of the link extending through the stock thickness in depth, or a distance in length equal to or greater than the stock thickness in the areas specified in Figure 1 as depicted in Figure 2.

6.4.2.2 Cracks - tears in nonprincipal areas. - Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length in all areas not specified in Figure 1 as depicted in Figure 3.

6.4.3 Burrs

6.4.3.1 Burrs in principal areas. - Any rough edge on any part of the link which, in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case or which would injure personnel in handling.

6.4.3.2 Burrs in nonprincipal areas. - Any rough edge on any part of the link that does not fall within the description of 6.4.3.1.

6.5 Changes from previous issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

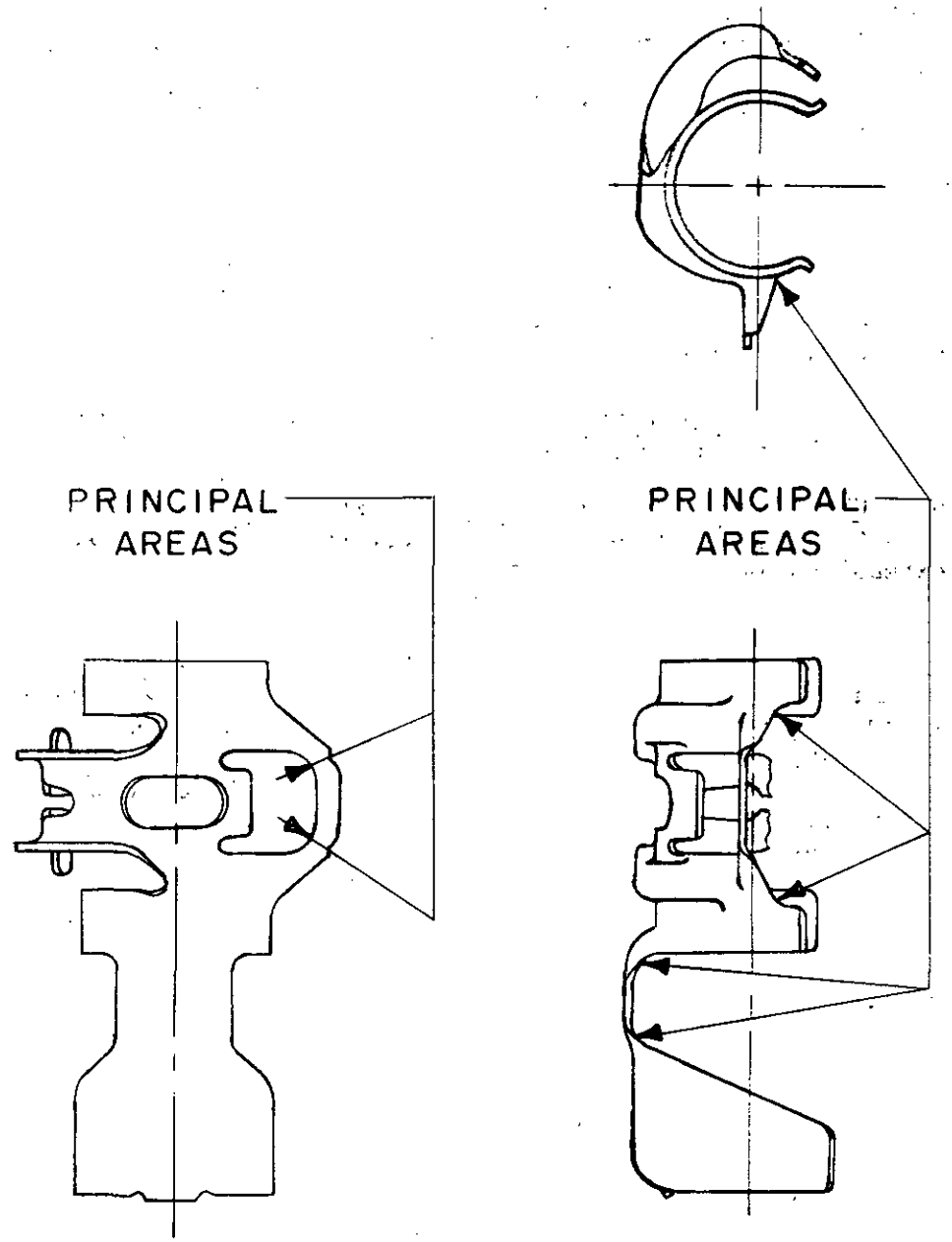
CUSTODIAN:
Army-AR
Air Force-99

REVIEW ACTIVITIES:
Air Force-99

PREPARING ACTIVITY
Army-AR

PROJECT NUMBER: 1305-0841

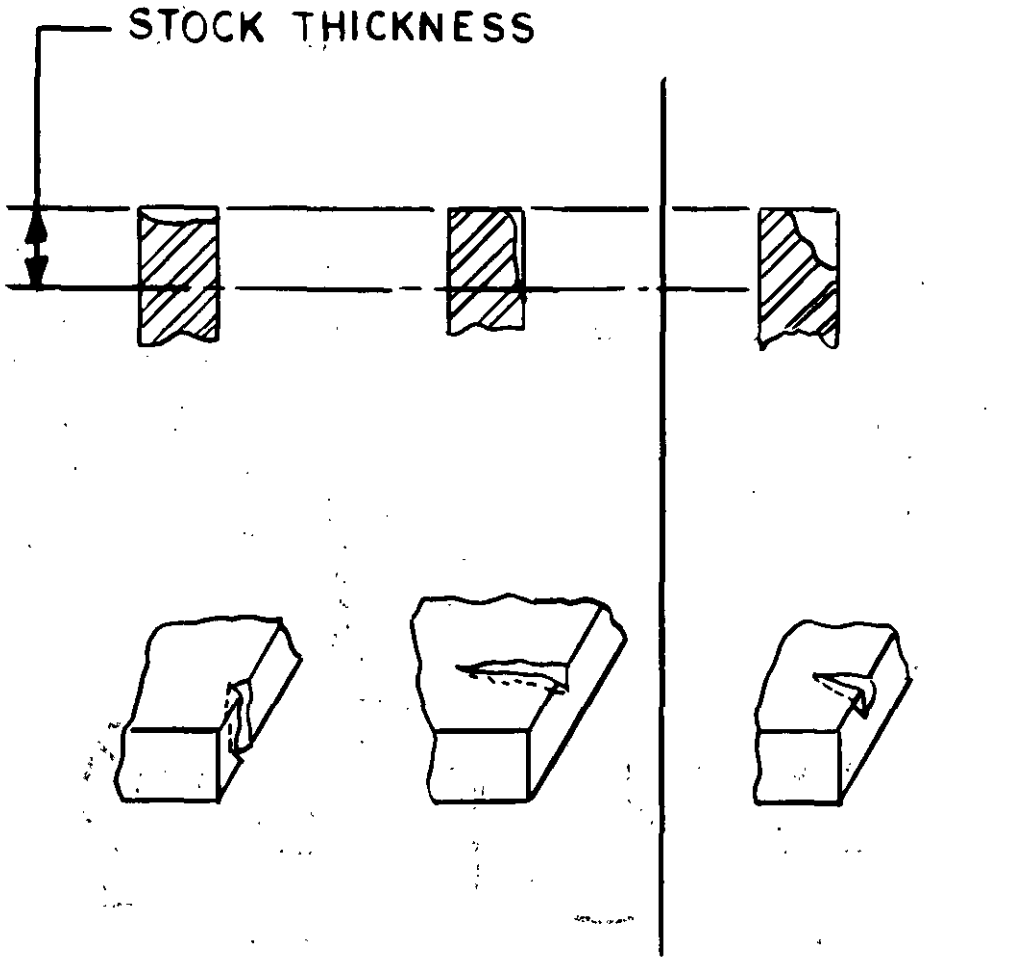
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.



LINK, CARTRIDGE, METALLIC BELT, 20MM: M22

FIGURE 1

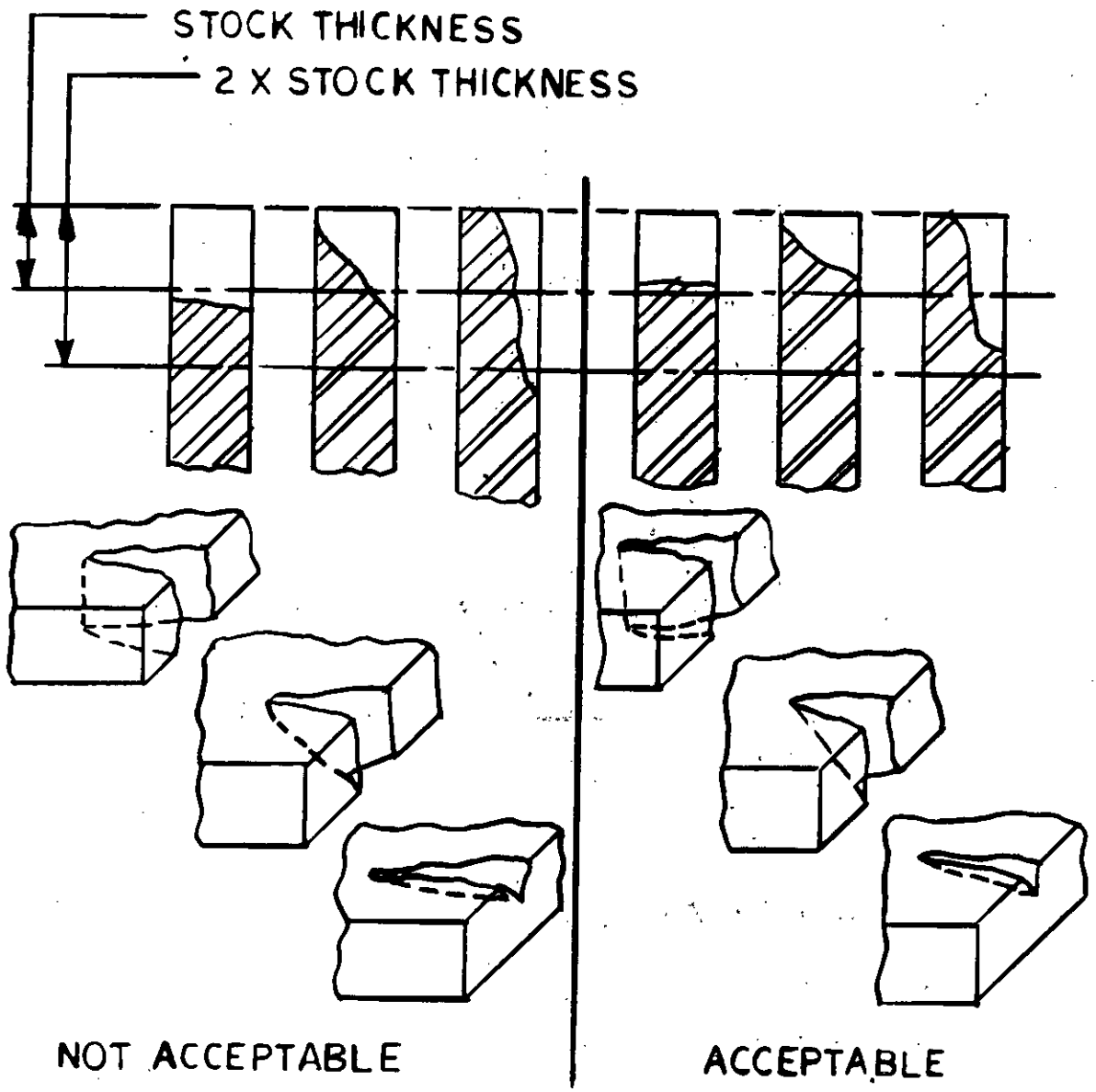
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.



CRACKS-TEARS IN PRINCIPAL AREAS

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.



CRACKS TEARS IN NONPRINCIPAL AREAS

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-L-45946A Link, Cartridge, Metallic Belt, 20MM: M22

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAA
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:25Z
Check the source to verify that this is the current version before use.